To complete this example, we will divide the s-domain voltage by the s-domain current, just as if we were using Ohm's law ($R = V/I$):

$$\frac{V(s)}{I(s)} = \frac{\dfrac{\omega_0 L s}{\omega_0^2 + s^2}}{\dfrac{\omega_0}{\omega_0^2 + s^2}} = sL$$

We find that the s-domain representation of the voltage across the inductor, divided by the s-domain representation of the current through the inductor, is equal to sL. This is *always* the case, regardless of the current waveform we start with. In a similar way, the ratio of s-domain voltage to s-domain current is always equal to R for resistors, and $1/sC$ for capacitors.

Figure 32-6 shows an example circuit we will analyze with the Laplace transform, the *RLC* notch filter discussed in Chapter 30. Since this analysis is the same for all electric circuits, we will outline it in steps.

Step 1. Transform each of the components into the s-domain. In other words, replace the value of each resistor with R, each inductor with sL, and each capacitor with $1/sC$. This is shown in Fig. 32-6.

Step 2: Find $H(s)$, the output divided by the input. As described in Chapter 30, this is done by treating each of the components as if they obey Ohm's law, with the "resistances" given by: R, sL, and $1/sC$. This allows us to use the standard equations for resistors in series, resistors in parallel, voltage dividers, etc. Treating the *RLC* circuit in this example as a voltage divider (just as in Chapter 30), $H(s)$ is found:

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{sL + 1/sC}{R + sL + 1/sC} = \frac{sL + 1/sC}{R + sL + 1/sC}\left[\frac{s}{s}\right] = \frac{Ls^2 + 1/C}{Ls^2 + Rs + 1/C}$$

As you recall from *Fourier* analysis, the frequency spectrum of the output signal divided by the frequency spectrum of the input signal is equal to the system's *frequency response*, given the symbol, $H(\omega)$. The above equation is an extension of this into the s-domain. The signal, $H(s)$, is called the system's **transfer function**, and is equal to the s-domain representation of the output signal divided by the s-domain representation of the input signal. Further, $H(s)$ is equal to the *Laplace transform* of the impulse response, just the same as $H(\omega)$ is equal to the *Fourier transform* of the impulse response.